ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #17

DjCraft v0.1.0

DJCraft is an ArtCraft-tier local-first 4-deck digital mixing, real-time stem extraction, and harmonic transition studio designed to run locally on Goliath without cloud latency or subscription paywalls.



v0.1.0	4 Crates	883	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\djcraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

DJCraft is an ArtCraft-tier local-first 4-deck digital mixing, real-time stem extraction, and harmonic transition studio designed to run locally on Goliath without cloud latency or subscription paywalls.

Features:

- **4-Deck Virtual Engine**: Direct fader control, pitch bends, transient-based BPM tracking, and master crossfader curves.
- **Real-Time Stem Isolation**: Dedicated stem gain multipliers for Vocals, Drums, Bass, and Other instrumental tracks.
- **Camelot Harmonic Wheel Intelligence**: Automated key clash prevention and harmonic compatibility algorithms ($8A \leftrightarrow 8B$, $8A \leftrightarrow 9A$, relative major/minor switches).
- **FastMCP Protocol**: Port `14118` interface for multi-agent autonomous performance swarms.

Workspace Architecture

1. `crates/djcraft-audio`: DSP abstractions, Camelot wheel logic, and stem isolation structures.

2. `crates/djcraft-core`: 4-deck orchestrator, harmonic transition planner, and track library indexing.

3. `crates/djcraft-mcp`: FastMCP JSON-RPC server on port `14118`.

4. `crates/djcraft-app`: Native GUI (`eframe`/`egui`) and CLI tool (`djcraft-cli`).

Verification & Usage

```
# Run test suite
cargo test

# Headless smoke test
./target/debug/djcraft.exe --smoke

# CLI Telemetry
./target/debug/djcraft-cli.exe status
./target/debug/djcraft-cli.exe transition
./target/debug/djcraft-cli.exe camelot 8A 9A
```

Status — does today / next week / non-goals

Does today: 4-crate Rust workspace (audio/core split), live-performance verbs declared, fleet i18n, 3 test fns.

Next week: PRD scope cut (live decks first, DAW/compose later) plus Effect/SoundCraft boundary; headless DSP fixture tests (mix → artifact + ms); transition-sim recipe.

Non-goals: not a DAW/arranger (studio side belongs to musiccraft); no cloud rendering.

Lineage — the Storytold craft-app pattern

Storytold ArtCraft (storytold/artcraft — "the IDE for artists") was the starting point. The 16 craft apps in the Storytold stable each ship one binary that is both the human CLI and the agent MCP server (stdio), headless-first, with `run.json` + report artifacts. That modus operandi and harness is what this repo implements; the code here is independent, the pattern is shared.